

Engineering Mechanics

Sem 2

Watermarked study resource

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CATEGORY: PYQs

STATUS: ok

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Note: ()

(1)

(For the candidates admitted from the academic year 2022-2023)

Part -A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed

over to hall invigilator at the end of 40th minute.

Part-B and Part -C should be answered in answer booklet

Time: 3 Hours

1. Mechanics is a branch of science which deals with

(A) Failure Analysis

(A) Force (C) Momentum

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() Study of bodies under rest only

2. Action of pull or push of a body at rest or motion is

8 kN force

Reg. No.

BTech. DEGREE EXAMINATION, MAY 2023 Second Semester

8 kN force

3 Two forces of magnitude 5 kN and 8 kN act vertically upwards and separated by 4m.

The resultant will be

(A) 3 kN at a distance of 2.46m from (B) 13 kN at the middle of two forces

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(C) Normal Force

(C) 13 kN at a distance of 2.46m from (D) 13 kN at a distance of 2.46m from 5

kN force

developed?

PART-A (20 x 1=20Marks) Answer ALL Questions

4. Varignon's theorem of moments states that if a number of coplanar forces acting on a particle are in equilibrium, then (A) Their algebraic sum is zero

5. Frictional force is independent of (A) Type of surface

(A) Dry friction (C) Fluid Friction

(C) The algebraic sum of their (D) moments about any point is equal to the moment of their resultant

force about the same point

raise the same load (A) Equal to (C) Less than

(B) Condition of rest or motion of bodies (D) Study of bodies under motion only

6. When there is no relative motion between contact surfaces a

(C) Joins only

8. The trusses are loaded at

(B) Moment (D) Work

(A) Mid point of the truss member

7. In a Screw Jack, the effort required to lower the load is

31 3

37 37 (C) $2r \sqrt{4r}$

(B) Their line of action are at equal distances

(B) (D)

The algebraic sum of their moments about any point in their plane is zero.

Area of contact Coefficient of friction

(B) Dynamic Friction (D) Static Friction

9. Centroid of a quadrant of a circle is given by (A) $T4-2r$

(B) More than (D) Un predictable

(D) Supports only

(B)

(B) Both joints and midpoint of the truss member

$Y = \frac{37}{37}$

31

the effort required to

force is

$2r \sqrt{37}$

Max. Marks: 75

Marks BL CO

1

1

1 1

PO

1

I 1 2

I 1 1

1 1 2 1

1 1 2 1

1 1 2 1

1 2 2 1

1 2 3 1

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(A) 15 cm (C) 2.5 cm

Radius of KYTHtion of a circular plate of diameter 10cm is

(C)

U. Moment of inertia of a sphere of radius R and mass M about an axis tangential to it is 2

(A)

(A) 4.5 (C) 13.5

(C)

2cm

12. Moment of Inertia of the rectangle about AB is

$2MR$

$2MR^3$

time $t=5$ seconds is

17.5

kN will be

210^2

(A) 1000 J (C) 1000 kJ

13. Motion of a particle along a straight line is given by $x = 4.5t + 5$, acceleration at

m

(A) Increases

(A) 24 (C) 60

3cm

B

(C) Remains Constant

(C) 600 J

(A) $2L$ (C) L

(B) $L.75\text{ cm}$ (D) 2.25 cm

2

(B)

(A) 5 (C) 15

14. Work done by a horse while pulling a cart for a distance of 1km exerting a force of 1

(D) $7MR$

(B) 9 (D) 18

20. A mechanism with five links has

(B)

(D)

16. In an elastic collision, the kinetic energy of the system

17. The circumferential velocity of 12m/s . A wheel rad/s of the wheel is

15. The centre of wheel rolling on a horizontal surface has a velocity of V_0 . $1\ 2\ 4\ 1$

A point on the rim that is level with the centre will move at a rate of

(A) V_0 (C) $V_2 V_0$

$7MR$

(B) $2V_0$ (D) V_0

(D)

30.0

(B) 100J (D) 100 kJ

5

23.52

3

(B) Decreases

18. Change in Kinetic energy of a rigid body due to linear motion and rotary motion are 400J and 800J respectively. The work done by the rigid body is

(A) 1200 J

(B)

(B) 6 (D) 15

(D)

19. A particle performs uniform circular motion with an angular momentum "L". If the frequency of the particles motion is doubled in and its kinetic energy halved, the angular momentum becomes

2

cm

First increases then reduces

(B) 400J

L L

4

of radius 50cm. The angular velocity 1 2 5 2

(B) 10 (D) 20

1 2 4 1

instantaneous centres

2 4 I

1 1 4 1

1 2 5 2

1 2 sI

1 1 5

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1

(D) 800 J

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21. a. Four forces of magnitudes 20N, 40N, 60N and 80N are acting respectively along the four sides of a square ABCD as shown. Locate the resultant force and its direction of action.

PART-B(Sx-40 Mark) Answer ALL Questions

ends.

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2m

60 N D

b. A simply supported beam is loaded as shown. Determine the support reactions at both

80 N

80 mm

(OR)

40 N 60 N

2m

60 mm

2m

23. a. A locate the centroid for the plane are shown.

(OR)

22. a. Derive an expression for the applied force "P" required to move the block up the plane.

120 mm

40 N

b. A ladder 4m long leans against a smooth vertical wall at an angle of 60° with the horizontal. The weight of the ladder is 700N. When a person of 600N stands on a rung 1.2m from the bottom of the ladder, the ladder is just about to slide. Determine the coefficient of friction between the ladder and the floor.

(OR)

162m

20 N

All dimensions are in "mm"

200

20 N

6m

$\tan \phi > \tan$

b. Determine the moment of inertia about the centroid axes for the T section shown.

60 mm 40 mm

[40

160

8 3 2 2

8 3 2 1

8 2 3 2

3 3 2

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Two vehicles travel between two stations 50km apart. They start at the same time and travel at 30kmph. If the first vehicle halts at the second station for 5 minutes and from the same station the first vehicle travels 50kmph whereas the second vehicle then starts with the same speed. Determine where the two vehicles will meet and at what time?

b. Three perfectly elastic balls A, B and C of masses 2kg, 6kg and 12 kg are moving in the same direction with velocities 12m/s, 4 m/s and 2 m/s respectively. If the ball A strikes ball B, which in turn strikes the ball C, Determine the velocity of the balls A, B & C after the collision.

25. a. Bar AB of length 5m slides in xy plane. The velocity of point A is 10m/s downwards and makes an angle 60° with the vertical. Determine the velocity of point B and mid-

Point C.

(OR)

b. A sphere, a cylinder and a hoop each having the same mass and radius are released from rest on an incline. Determine the velocity of each body after it has rolled through

(OR)

a distance corresponding to a change in elevation "h"

3 m

PART-C(1x15 = 15 Marks) Answer ANY ONE Questions 26. Find the forces in all members of the truss as shown.

A 200 N

B

B

36kN 3 m

27. Find the acceleration of the blocks and tension in the cable for the system shown.

DI

B150 N

Merta

1

4 S

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